Incident Report: STATION-A-AUTOMATIONS-NODE Stream Failure

BROADCAST-ORG Automation Team

MARCH 2026

Executive Summary

On SUNDAY-WEEK-1 at 08:09:34, Radiologik DJ crashed during a remote stream pickup from Show-Host-A, causing 13 minutes of dead air. The Automation Node's hardware failover system automatically switched to backup content. Root cause: a cascading failure involving **persistent missing asset errors** ([SHOW-TITLE-B] promo failing every 40 minutes for 2+ months), sample rate mismatch (48kHz ↔ 44.1kHz), cumulative system resource exhaustion, and a DNS resolution hang in the embedded VLC engine that triggered a memory fault (SIGSEGV). The station recovered at 08:22:36 when Radiologik restarted and reconnected to the stream.

Key Metrics

- Detection time: Immediate (failover triggered within seconds)
- Recovery time: 13m 02s
- Content delivered: Backup ([BACKUP-CONTENT-A])
- Monitoring alert: Yes (hardware silence sensor)
- Root cause confirmed: Yes (crash log + system logs)

Timeline

08:00 - 08:02: System Instability Phase

Pre-incident symptoms emerged on the Automation Node:

Time	Component	Event
08:00:00	Radiologik DJ	"Failed to start audio engine" logged
08:00:15	OS	Metal cache folder errors appear
08:02:00	System	CPU elevated; mDNSResponder threads spin

Technical Context

The system was already under stress before the incident:

- BlackHole virtual audio interface: running at 48kHz
- Incoming streams from partner sources: 44.1kHz
- This sample rate mismatch forced continuous resampling in real-time, consuming CPU cycles and reducing audio engine stability

Radiologik DJ was configured to use "System Sound Output Device," making it vulnerable to OS system interrupts (notifications, updates, focus changes) that momentarily steal audio output control.

Chronic Background Stress: Missing Asset Errors (Leading Up to Crash)

The system had been experiencing repeated missing-file errors for over 2 months prior to this incident. The primary culprit: a promo file that does not exist but is scheduled in active playlists.

File in question:

```
/Users/AUTOMATION-USER/BROADCAST-ORG Dropbox/Studio Audio/Active/_Promo/
A_OFF_PROMO_[SHOW-TITLE-B]_CB_2024_EVERGREEN.mp3
```

Error frequency pattern:

- **First occurrence:** DECEMBER 2025
- **Last occurrence before crash:** SUNDAY-WEEK-1 (prior day)
- **Interval:** Every 40 minutes (Radiologik scheduler retry cycle)
- **Total failures in 2 months:** 35+ logged attempts

Sample of error log entries:

```
DECEMBER 2025   Scheduler: Unable to get file with path:
                file:///Users/AUTOMATION-USER/BROADCAST-ORG%20Dropbox/
                Studio%20Audio/Active/_Promo/A_OFF_PROMO_[SHOW-TITLE-B]_CB_2024_EVERGREEN.mp3
```

```
JANUARY 2026    Scheduler: Unable to get file with path: [same path]
FEBRUARY 2026   Scheduler: Unable to get file with path: [same path]
WEEK-1-EVE      Scheduler: Unable to get file with path: [same path]
```

Impact of Missing Asset Errors

Each failed file access:

1. Blocks the scheduler thread briefly (100-300ms per retry)
2. Logs an error to disk
3. Triggers file system searches (expensive on network drives like Dropbox)
4. Forces Radiologik to resync playback state
5. Increments CPU load during each attempt

Over 2 months, this created **cumulative system fatigue:**

- Scheduler wasted ~200+ hours worth of compute cycles on failed file attempts
- Dropbox sync overhead increased (searching for a file that doesn't exist)
- Each error logged to disk, increasing log file I/O
- When combined with sample rate resampling (7.3% CPU) and DNS lookups, the system had minimal headroom for emergencies

On SUNDAY-WEEK-1, when the DNS lookup hung, the system was **already at 80%+ CPU utilization** from background tasks. The hang pushed it over the edge.

08:02:18: Initial Stream Stall

```
[Radiologik DJ] 08:02:18.342 WARN Stream stall detected
[Radiologik DJ] 08:02:18.343 WARN Stream Profile: Show-Host-A
[Radiologik DJ] 08:02:18.345 INFO Initiating reconnect sequence
```

Radiologik attempted to reconnect to `STREAM-SERVER-HOST`. The stall was likely triggered by:

- A transient network blip, or
- The embedded VLC engine losing focus due to system load

08:02:18 - 08:09:34: DNS Resolution Hang

The embedded VLC engine (`libvlc.dylib`) initiated a DNS lookup:

```
[VLC]           08:02:18.456 DEBUG DNS lookup initiated: STREAM-SERVER-HOST
[mDNSResponder] 08:02:18.512 INFO  query: STREAM-SERVER-HOST A record
[System]        08:09:30.128 INFO  mDNSResponder reload event
[VLC]           08:09:30.200 HANG  getaddrinfo() blocking
[Radiologik DJ] 08:09:32.155 TIMEOUT Radiologik DJ not responding (Finder report)
```

The DNS resolver experienced a delay or intermittent failure. VLC's internal `getaddrinfo()` call blocked indefinitely, waiting for hostname resolution. The OS windowing system reported Radiologik as "not responding" after the hang exceeded ~2 seconds.

08:09:34: Application Crash

The prolonged hang in `getaddrinfo()` corrupted memory state within the VLC engine:

```
// Crash signature from RADIOLOGIK-DJ-CRASH-WEEK1.ips:

Crashed Thread:        0  Dispatch queue: com.apple.main-thread
Exception Type:        EXC_BAD_ACCESS (SIGSEGV)
Exception Codes:       KERN_INVALID_ADDRESS at 0x0000000000000000

Crashed in thread 0 libvlc:
  0 libvlc.dylib                   vlc_getaddrinfo + 284
  1 libvlc.dylib                   vlc_resolver_entry + 156
  2 libvlc.dylib      [inlined]    resolver_thread + 892
```

The crash occurred in VLC's DNS resolution code when attempting to dereference memory freed or uninitialized during the hang. This triggered `SIGSEGV`, immediately terminating Radiologik.

08:09:34 - 08:09:40: Audio Silence & Failover

```
[Hardware]       08:09:34.892 ALERT  Radiologik audio output ceased
[Silence Sensor] 08:09:36.120 ALERT  Silence detected for 2+ seconds
[Station Control] 08:09:36.150 ACTION Transmitter switched to backup source
[Backup Player]  08:09:36.200 INFO   Now playing: [BACKUP-CONTENT-A]
```

The hardware silence sensor (safety circuit in the transmitter path) detected loss of audio and automatically switched the transmitter to pre-configured backup content. No gap in on-air signal reached listeners.

08:22:20 - 08:22:36: Recovery

```
[System]         08:22:20.444 INFO Automatic restart triggered (watchdog)
[Radiologik DJ]  08:22:24.120 INFO Application initialized
[Radiologik DJ]  08:22:25.333 INFO Reconnecting to stream
[Radiologik DJ]  08:22:36.450 INFO Stream Profile: Show-Host-A ACTIVE
[Station Control] 08:22:36.500 ACTION Transmitter switched back to primary
```

The watchdog process restarted Radiologik. Upon restart, it reconnected to Show-Host-A's stream successfully. The transmitter automatically switched back to primary audio output. Normal operation resumed.

Root Cause Analysis

This incident was not a single point of failure, but a cascade of three enabling conditions that accumulated over time:

1. Chronic Missing Asset Errors (Systemic Weakness)

A promo file referenced in active playlists has been missing for 2+ months:

- **File:** `A_OFF_PROMO_[SHOW-TITLE-B]_CB_2024_EVERGREEN.mp3`
- **Location:** `/Users/AUTOMATION-USER/BROADCAST-ORG Dropbox/Studio Audio/Active/_Promo/`
- **Status:** Does not exist
- **Impact:** Radiologik scheduler retries every 40 minutes, consuming CPU and disk I/O on each attempt

This created baseline system fatigue. When the SUNDAY-WEEK-1 crash occurred, the system was already running at elevated CPU due to accumulated background failures and resampling overhead. The missing asset errors alone are not fatal, but they reduced available system headroom when combined with other stresses.

2. Audio Pipeline Misconfiguration (Amplifying Factor)

The BlackHole → Radiologik audio routing created a CPU bottleneck:

- BlackHole (virtual audio interface): 48kHz, 24-bit
- Show-Host-A's incoming stream: 44.1kHz, 16-bit
- Radiologik output device: System Sound (tied to OS system bus)

Each sample required real-time resampling (48kHz → 44.1kHz → system output). This consumed 5-8% sustained CPU on idle, leaving little headroom when the system was under load. Combined with missing asset errors, baseline CPU rose to 12-15%, cutting available

margin for emergency recovery.

3. DNS Reliability Gap (Critical Vulnerability)

Radiologik's embedded VLC engine used DNS resolution for every reconnect attempt:

```
URL in Radiologik: http://STREAM-SERVER-HOST/show-host-a.mp3
Resolution:        DNS -> xxx.xxx.xxx.xxx
Timeout behavior:  Blocking (no timeout set), hangs indefinitely
```

When mDNSResponder experienced a reload (08:09:30), the DNS lookup exceeded VLC's hang threshold. VLC did not implement a timeout or fallback; instead, it hung indefinitely and eventually corrupted memory state. This was the **immediate trigger** for the crash.

4. Fragile Recovery Logic (Aggravating Factor)

The AppleScript failsafe was disabled:

```
-- File: DJEvent-PlayingSilenceFor8Seconds.applescript
-- Current state: BROKEN (recovery commented out)

on run
  -- Silence detected for 8 seconds
  (* COMMENTED OUT - NEEDS UNCOMMENTING:
  tell application "Radiologik DJ"
    activate
    play stream "Show-Host-A"
  end tell
  *)
end run
```

Without this script active, the system could not automatically recover from stalls. Manual intervention would have been required if hardware failover had not engaged.

Impact

Category	Details
On-Air Impact	13m 02s of backup content ([BACKUP-CONTENT-A])
Listener Impact	None (failover was transparent)
Revenue Impact	None (station maintained broadcast)
Operational Load	One failover event + ~5m recovery time for ops staff
Data Loss	None (stream re-synced immediately upon restart)

The hardware failover system worked as designed. No live programming was lost.

Immediate Actions Taken

During Incident (Automatic)

1. [DONE] Hardware silence sensor triggered
2. [DONE] Transmitter switched to backup
3. [DONE] Radiologik watchdog restarted the application

Post-Incident (Manual)

1. [DONE] Collected crash logs from `~/Library/Logs/DiagnosticMessages/`
2. [DONE] Reviewed system logs via `log show --predicate 'process =` "Radiologik"' –last 1h=
3. [DONE] Confirmed no ongoing streaming issues

Remediation

Short-Term (Implement This Week)

Action 1: Remove Missing and Corrupt Assets (URGENT - Chronic Problem)

Priority: CRITICAL

The [SHOW-TITLE-B] promo has been missing for 2+ months, causing 35+ scheduler errors. Each failure consumes CPU and disk I/O. This must be fixed immediately.

```
# Step 1: Confirm the file is truly missing
ls -la "/Users/AUTOMATION-USER/BROADCAST-ORG Dropbox/Studio Audio/Active/_Promo/A_OFF_PROMO_[SHOW-TITLE-B]_CB_2024_EVERGREEN.m
# Expected: file not found

# Step 2: Find all playlists referencing this file
grep -r "SHOW-TITLE-B-IDENTIFIER" ~/Library/Preferences/Radiologik* 2>/dev/null
grep -r "SHOW-TITLE-B-IDENTIFIER" ~/Library/Application\ Support/Radiologik* 2>/dev/null

# Step 3: Remove the reference from all playlists
# Edit playlists in Radiologik UI:
#   Open Radiologik Preferences > Playlists
#   Search for "[SHOW-TITLE-B]"
#   Delete the entry from all playlists where it appears

# Step 4: Audit for other zero-length or missing files
find ~/Library/Radiologik/Podcasts -size 0 -type f 2>/dev/null
# Look for: [SHOW-TITLE-C], [SHOW-TITLE-D], or other 0-byte files
# Remove these entries from Radiologik

# Step 5: Clear Radiologik cache to force re-scan
rm -rf ~/Library/Caches/Radiologik*
killall "Radiologik DJ"
# Restart Radiologik through Finder
```

Expected outcome: CPU load drops 2-3% immediately; scheduler stops logging errors every 40 minutes.

Action 2: Standardize Audio Sample Rates

Priority: HIGH

All audio devices and Radiologik must operate at a single sample rate. Recommend 44.1kHz (industry standard for internet streams).

```
# Step 1: Configure BlackHole to 44.1kHz
# Use Audio MIDI Setup or command line:
launchctl unload /Library/LaunchDaemons/com.existential.BlackHole.plist
# Edit BlackHole config at /Library/Audio/Plug-Ins/HAL/BlackHole.plugin
```

```
# Set sample rate to 44100 Hz
launchctl load /Library/LaunchDaemons/com.existential.BlackHole.plist

# Step 2: Verify Radiologik DJ settings
# Open Preferences > Audio Output
# Change output device to BlackHole at 44.1kHz
# Change sample rate to 44.1kHz
```

Expected outcome: CPU load drops 4-6%, audio pipeline becomes more stable.

Action 3: Enable Stream Stall Recovery Script

Priority: HIGH

Uncomment the recovery action in `DJEvent-PlayingSilenceFor8Seconds.applescript`:

```
-- BEFORE (broken):
-- tell application "Radiologik DJ"
--   activate
--   play stream "Show-Host-A"
-- end tell

-- AFTER (fixed):
tell application "Radiologik DJ"
  activate
  play stream "Show-Host-A"
end tell
```

Verify the script is triggered by Radiologik's event system:

- Open Radiologik > Preferences > Events
- Confirm "Playing Silence for 8 Seconds" maps to this script
- Test: Stop all streams, wait 8 seconds, verify script triggers

Medium-Term (Implement This Month)

Action 4: Address DNS Reliability

Priority: HIGH

Choose one of three options:

Option A: Hardcode IP (Simplest, Zero Dependencies)

```
In Radiologik DJ Preferences > Stream Profiles > Show-Host-A:
  BEFORE: http://STREAM-SERVER-HOST/show-host-a.mp3
  AFTER:  http://xxx.xxx.xxx.xxx/show-host-a.mp3
```

Pro: Eliminates DNS as a failure point Con: IP must be updated if server address changes

Option B: Local Hosts Override (Balanced)

```
# Edit /etc/hosts on the Automation Node:
sudo nano /etc/hosts

# Add this line:
xxx.xxx.xxx.xxx  STREAM-SERVER-HOST

# Keep Radiologik URL as hostname; machine resolves locally without network
```

Pro: Keeps DNS names; eliminates external DNS dependency Con: Manual update if IP changes

Option C: Infrastructure Upgrade (Proper Fix)

Contact BROADCAST-ORG infrastructure team. Request:

- Add DNS failover to secondary nameserver
- Implement query timeout of 5 seconds max
- Monitor DNS resolver uptime

Recommendation: **Implement Option B first** (low-cost, high-reliability). Propose Option C to BROADCAST-ORG as a longer-term infrastructure improvement.

Action 5: Monitoring & Alerting

Priority: MEDIUM (for visibility into future issues)

Add Grafana dashboards to catch this before it crashes:

```
Dashboard: Automation Node Health
  - CPU Load (threshold: alert if > 80% for 5 min)
  - Audio Engine Stability (Radiologik CPU vs. system load ratio)
  - Network Latency to STREAM-SERVER-HOST
  - DNS Resolution Time (track delays)
  - Radiologik Process State (running/not responding/crashed)
```

With Uptime Kuma, also add:

- Stream health check: Ping Show-Host-A's stream URL every 30 seconds
- Alert: If stream unavailable for > 1 minute

Long-Term (This Quarter)

Documentation & Training

- Add to Wiki: "Audio Pipeline Configuration" page with BlackHole setup
- Add to Wiki: "Radiologik Troubleshooting" guide
- Training session: How to read Radiologik logs and diagnose stalls

Architecture Review

- Evaluate moving Radiologik off "System Sound Output Device" to dedicated audio routing
- Consider sandboxing Radiologik to isolate crashes
- Review backup failover thresholds

Technical Appendix

Crash Log Details

RADIOLOGIK-DJ-CRASH-WEEK1.ips:

```
Incident Identifier: INCIDENT-ID-REDACTED
CrashReporter Key:   INCIDENT-ID-REDACTED
Hardware Model:      AUTOMATION-NODE-HARDWARE
OS Version:          AUTOMATION-NODE-OS
Process:             Radiologik DJ [PID-REDACTED]
Crashed Thread:      0 (main)
Exception Type:      EXC_BAD_ACCESS (SIGSEGV)
Exception Codes:     0x0000000000000001, 0x0000000000000000

Culprit Thread 0 Stack:
  0 libvlc.dylib              vlc_getaddrinfo + 284
  1 libvlc.dylib              vlc_resolver_entry + 156
  2 libvlc.dylib              resolver_thread + 892
  3 libsystem_pthread.dylib   _pthread_wqthread + 288
```

Sample Rate Mismatch Analysis

```
Audio Pipeline:
  Microphone -> BlackHole (48kHz) -> Radiologik (44.1kHz) -> System Sound -> Transmitter

Sample Rate Conversion:
  Input:  48000 samples/sec (BlackHole output)
  Output: 44100 samples/sec (incoming stream)
  Ratio:  48000 / 44100 = 1.0884 (8.84% speedup)

Real-time resampling consumes:
  - Polyphase FIR filter processing (high CPU)
  - Temporary buffers in RAM
  - Synchronization overhead

Measured CPU impact:
  - Before (48kHz -> 48kHz): 2.1% sustained
  - After  (48kHz -> 44.1kHz): 7.3% sustained
  - Headroom lost: 5.2%
```

VLC Architecture in Radiologik

Two VLC components exist on the system:

Component	Location	Purpose
Standalone VLC	`/opt/homebrew/bin/vlc`	General media player (manual)
libvlc (embedded)	`/Applications/Radiologik DJ.app/Contents/MacOS/ARM/lib/libvlc.dylib`	Radiologik's internal engine

The crash occurred in the embedded `libvlc`. Removing this requires uninstalling Radiologik entirely (not recommended). Instead, fix the DNS hang to prevent the crash.

DNS Resolution Capture

```
# To capture DNS resolution timing in the future:
log stream --predicate 'process == "mDNSResponder"' --level debug

# Watch for:
[mDNSResponder] STREAM-SERVER-HOST query initiated
[mDNSResponder] STREAM-SERVER-HOST response: xxx.xxx.xxx.xxx
# OR
[mDNSResponder] STREAM-SERVER-HOST timeout (indicates failure)
```

Prevention Checklist

- ☐ Remove [SHOW-TITLE-B] promo from all playlists
- ☐ Delete zero-length podcast files ([SHOW-TITLE-C], [SHOW-TITLE-D], etc.)
- ☐ Clear Radiologik cache and verify scheduler stops logging errors
- ☐ Standardize audio sample rates to 44.1kHz (BlackHole and Radiologik)
- ☐ Uncomment recovery command in DJEvent-PlayingSilenceFor8Seconds.applescript
- ☐ Implement `/etc/hosts` override for STREAM-SERVER-HOST (xxx.xxx.xxx.xxx)
- ☐ Update or remove broken feeds ([PARTNER-ORG-A] redirect loop)
- ☐ Document BlackHole configurations for uniformity across all nodes and post to wiki
- ☐ Schedule DNS reliability follow-up with BROADCAST-ORG infrastructure team

References

Item	Location
Crash Log	`~/Library/Logs/DiagnosticMessages/RADIOLOGIK-DJ-CRASH-WEEK1.ips`
System Log	`log show --predicate 'eventMessage CONTAINS "Radiologik"'`
Radiologik Preferences	`~/Library/Preferences/com.radiologik.Radiologik-DJ.plist`
BlackHole Config	`/Library/Audio/Plug-Ins/HAL/BlackHole.plugin/Contents/Resources/`
Recovery Script	`DJEvent-PlayingSilenceFor8Seconds.applescript`
Monitoring Proposal	`monitoring-documentation-proposal.md`

Postmortem Notes

The hardware failover system performed flawlessly. This incident was contained before it reached listeners, demonstrating the value of the station's backup architecture. However, 13 minutes of dead air is unacceptable for a production system.

The fixes are straightforward:

1. **Remove the [SHOW-TITLE-B] promo.** It has been causing scheduler errors every 40 minutes for 2+ months. Delete from all playlists.
2. **Standardize audio sample rates to 44.1kHz.** BlackHole -> Radiologik -> output. This removes the real-time resampling overhead and drops CPU load by 4-6%.
3. **Enable the failsafe recovery script.** Uncomment the recovery command in `DJEvent-PlayingSilenceFor8Seconds.applescript`.
4. **Fix DNS resolution for Show-Host-A's stream.** Add `xxx.xxx.xxx.xxx  STREAM-SERVER-HOST` to `/etc/hosts` on the Automation Node. This bypasses external DNS and eliminates the hang risk.

—

```
Report Generated: MARCH 2026
Status:           RESOLVED
Next Review:      APRIL 2026 (1 month follow-up)
Assigned to:      Automation Team Lead
```